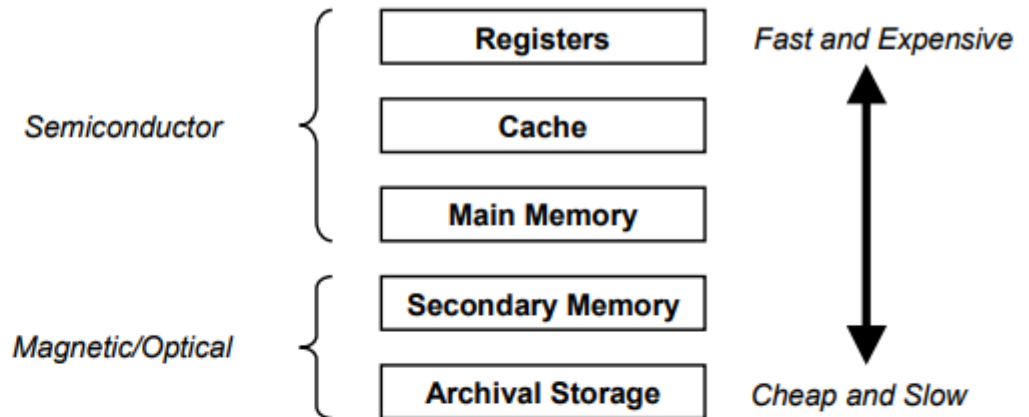


Memory types

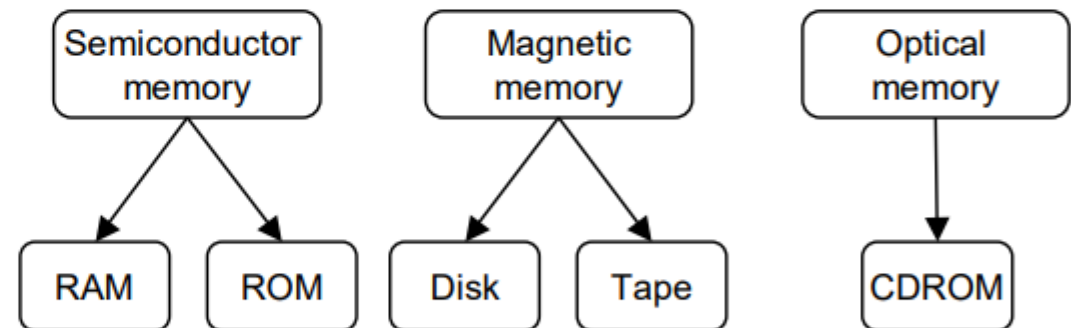
■ Classification by access

- **Registers:** Internal to the CPU
- **Cache:** Tightly linked to the CPU
- **Main memory:** Connected to system buses
- **Secondary memory:** On-line
- **Archival:** Off-line



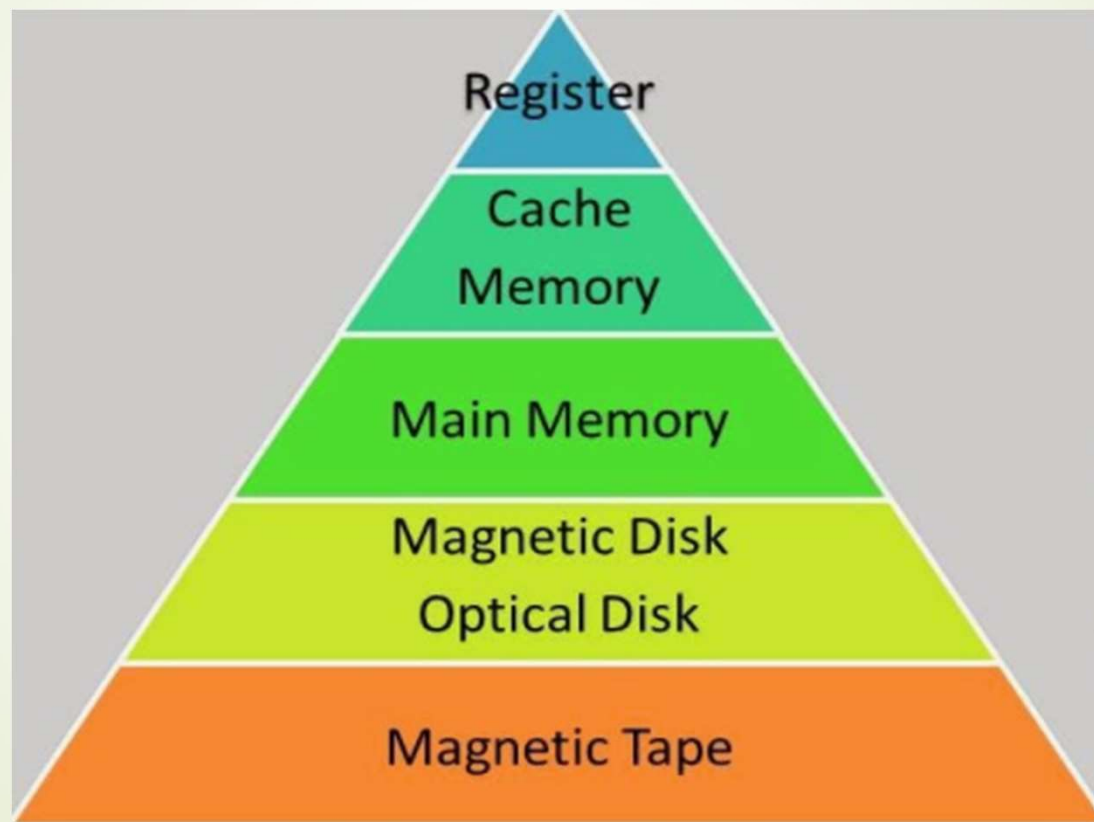
■ Classified by technology

- Semiconductor
 - Random Access Memory (RAM)
 - Volatile, Read and write
 - Read Only Memory (ROM)
 - Non-volatile, Can only be read
- Magnetic memory
 - Disk, Tape
- Optical
 - CD-ROM



MEMORY ACCESS	MEMORY TECHNOLOGY				
	RAM	ROM	Disk	Tape	Optical
Registers	X				
Cache	X				
Main memory	X	X	Virtual		
Secondary memory			X		X
Archival storage				X	X

Hierarchical memory system





Contd...

- The design constraints on computer memory depends on:
 - capacity
 - speed / access time
 - cost
- However, there is a trade-off among these three characteristics and the following relationship is observed.
 - => lesser access time, greater cost per bit.
 - => greater capacity, smaller cost per bit.
 - => greater capacity, greater access time.
- Desire=> greater capacity, faster access time, low cost per bit.
- Cannot be achieved with single memory component, solution is to organize computer memory into hierarchy.
- While moving down the hierarchy, we can observe the following:
 - => Increasing capacity.
 - => Decreasing cost per bit.
 - => Increasing access time / Decreasing speed.